Dependencies of the proofs in Topic C

Human Compression Project

This document explains the paired mathematical graph for the published common paper and its four applications. The graph preserves the forty source clauses already admitted with that edition. It records which results are used to prove other results; it does not assign costs, claim new results, or measure compression. All scientific files are the exact files published at commit `87757bf`.

1 The common arguments and their applications

A connected graph on a set Ω gives generators $e_u + e_v$ for the even finite-support binary module $E(\Omega)$. Products around closed walks give the relations. Choosing paths from a base vertex constructs an inverse to this presentation. If a group H acts on the graph, its conjugates of a suitable set of edge differences consequently generate $E(\Omega) \rtimes H$. This is the common generation argument used by the generalized cubes and the polyhedra with equal adjacent parameters. The cube application still retains Danzer's construction and its full automorphism-group theorem.

For the latter polyhedra, the separate recognition theorem constructs the inverse $(v, h) \mapsto T(v)j(h)$ from actual translation and group homomorphisms. Covariance makes this map multiplicative, and the explicit generation hypothesis makes it onto. Cyclic displacement commutativity, attributed to Li–Zhang and proved in full, verifies the local hypothesis. The application obtains the universal presentation from two inverse maps. Its proof therefore does not use the historical finite-order upper bound. The group intersection property, the matrix change of basis and the polytope correspondence remain explicit local obligations. The standard last correspondence is a retained theorem.

The geometric operation arguments have a different common starting point. Typed coset Helly supplies flag completion, while marked subgroup intersections supply the flag stabilizers. Their roles are distinct. The retained-object lemma computes the kernel of an operation's action from an equivariant encoding of its objects. Graph halving and geometric Petrials both use that lemma. Its use for Petrials can give a nontrivial kernel, so faithfulness is proved under the later thinness hypothesis rather than presumed at the outset.

Endpoint replacement is governed by an explicit mixed-parabolic intersection and a least-support condition on a finite index set. In the odd-order case the new markers recover the old group. The Janko application applies this criterion to its actual subgroup intersections, then checks typed Helly separately. The complete classification uses the extension-orbit lemma on proper residues. Its two full GAP programs remain mathematical dependencies: one checks the literal constructions and their presentation, and one exhausts the extension census. The operational driver is not a third mathematical program.

2 Historical dependencies are also recorded

Hartley–Leemans already invoke the Aschbacher recognition theorem, an earlier Janko presentation and subgroup data. Several later operation proofs explicitly refer to the argument of Theorem 4.2 or are left as exercises using preceding constructions. Those source arrows are present in the graph. Its introductory exhaustive classification retains the reported earlier MAGMA census; the original implementation and its cost are not silently replaced by zero.

Montero–Ivić Weiss use the quoted criterion for maximal parabolic intersections, a lemma on direct products and regular section groups for their halving intersection proof. Their flag-transitivity proof invokes chain transitivity in a semiregular polytope. The current graph-halving proof has a different route through actual objects and residues. The graph records the printed historical argument without endorsing the opening common-representative choice as an independent proof of flag transitivity. The original application retains Danzer and the classical Coxeter classifications. Its use of Ens's theorem is distinguished from the current application's more limited use of the lattice notation: the latter proves its own face and marking maps.

Bonheure–Buekenhout–Leemans use connected truncations, geometric Petrie walks and successive automorphism inclusions. The unconditional inclusion as an embedding is false for deduplicated geometric Petrie faces. It is flagged in the historical graph, including its use in the printed proof of automorphism equality. The current proof computes the kernel first and establishes faithful action under thinness. The geometric length assertion is also flagged; the stated counterexample has step order six and only three vertices in its Petrie face. The original PDF is unavailable. Exact browser-display lines locate the source body, but provide no replacement for its unknown manuscript cost.

Li–Zhang's matrix powers and translation basis feed their generation and intersection proofs. Their universal presentation proof uses displacement commutativity and a finite coset upper bound, then derives reversal and the parameter corollaries. These are actual source dependencies, including standard retained polytope theory. The inconsistent inverse-action multiplication display is visibly qualified, not accepted as the law of a left semidirect product. The higher-rank conjecture is outside this graph's target.

3 Fixed scope and evidence

The correspondence has eleven Janko rows, sixteen generalized-cube rows, eight Petrial rows and five equal-parameter rows. N14, the comparison against every example in a cited catalogue, and N15, the unused semiregular identification, are excluded auxiliary routes. They have no proved replacement output. J10 retains the correction to the two source figure labels, N13 the rank qualification, and P4, P7 and P8 their exact corrected or qualified statuses. F2 separately records the multiplication convention, and F5 treats $n = 2$ independently. This is not a claim of forty unqualified recoveries.

The accepted common-paper context binds both the graph geometry and its action adapters. For an acyclic logical graph these are refined into three actual arguments: flag extension and residues; faithfulness from the retained objects; and the subsequent identification of panel generators and parity support. The first two assemble the full graph-halving theorem. The third uses that completed theorem. This refinement changes no mathematics or source boundary.

Every node has exact line and codepoint locators and hashes. The forty accepted source records are preserved verbatim. Logical arrows point from prerequisite to use. Attribution, comparison, program replay and Lean evidence have separate edge types and do not become mathematical premises merely through traversal. The scoped Lean evidence covers sixteen declarations in three modules. It does not formalize the complete geometric or source-specific proofs.

The additive revision records six findings from the independent semantic read. It restores the Janko method's seed, generation and subgroup premises; the tail-triangle premise of the historical flag-transitivity route; and the retained Cayley–Hamilton and tetrahedral inputs. It also removes an unused binary-module premise from the purely group-theoretic displacement proof and restores the geometric premise for the incidental Janko residue names. Neither the scientific texts nor the forty-row scope has changed.

The optimized Python check verifies all bindings, closures and fifteen corruptions. These checks establish structural consistency, not the truth or minimality of an edge. The graph is an author proposal awaiting independent semantic review. Its overlapping proof locators are unweighted. An exclusive cost partition is separate work; node counts do not measure compression.